

Frequentist vs. Bayesian Inference

Presentation Outline

1 Frequentist Statistics

2 Bayesian Statistics

Frequentist Inference - Parameters

- Fixed but Unknown Values
- Hypothesis Tests for Parameters
 - p -value = probability the sample data or more extreme was produced from the model with the fixed parameter value as stated in the null hypothesis.
 - Provides evidence against null hypothesis value of parameter.
 - No statement of probability null hypothesis is true.

Bayesian Statistics - Parameters

- Random Variables with Distribution
- Before Data Collection - Prior Distribution for Parameter
- Data updates Prior Distribution to Posterior Distribution
- Can make probability statements about parameter's Value